

JEE ADVANCED - 10 | PAPER - 1

JEE 2022

Date	Maximum Marks	Duration
13th March, 2022	183	3 Hours

General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **Three** sections (Section 1, Section 2 & Section 3).
- Section 1** contains **7 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
- Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Section 3** contains **2 Tables** (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME**SECTION - 1**

- This section contains **SEVEN (07)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If only (all) the correct option(s) is(are) chosen
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct
Partial Marks: +1 If two or more options are correct but **ONLY** one option is chosen, and it is a correct option
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -2 In all other cases.

SECTION – 2

- This section has Five **(05)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If **ONLY** the correct numerical value is entered.
Zero Marks: 0 In all other cases.

SECTION – 3

- This Section has **2 Tables** (each having 3 columns and 4 rows). Based on each table, there are **Three (03) Matrix Match Type** questions. Each question has **FOUR** options out of which **ONLY ONE** is correct.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If **ONLY** the correct option is chosen.
Zero Marks: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -1 In all other cases.

SECTION 1

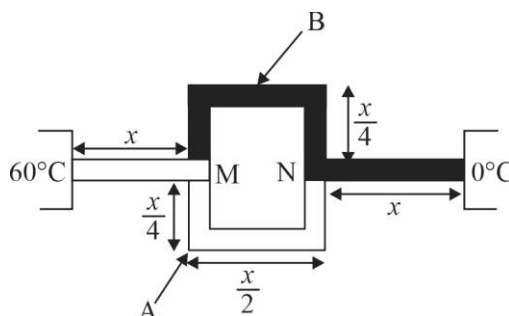
MULTIPLE CORRECT ANSWERS TYPE

This section consists of SEVEN (07) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. Tension in a uniform string varies with length such that tension at one end is T_0 and at other end $15 T_0$. If length of the string is L and mass of string is m_0 then time taken by pulse while moving from one end to other is :

(A) $\sqrt{\frac{T_0}{m_0 L}}(\sqrt{15}-1)$ (B) $\sqrt{\frac{m_0 L}{T_0}}(\sqrt{15}-1)$ (C) $\sqrt{\frac{m_0 L}{4T_0}}(\sqrt{15}-1)$ (D) $\sqrt{\frac{m_0 L}{49T_0}}(\sqrt{15}-1)$

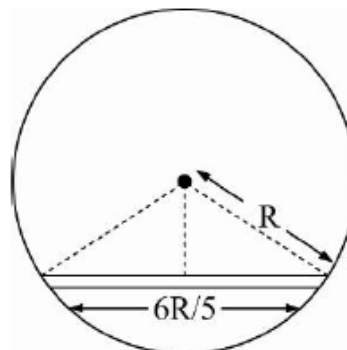
2. Thin rods made of material A and B having thermal conductivity k and $2k$ respectively are arranged between two reservoirs at 60°C and 0°C as shown in figure. The temperature of the junctions at M and N will be (approximately) :



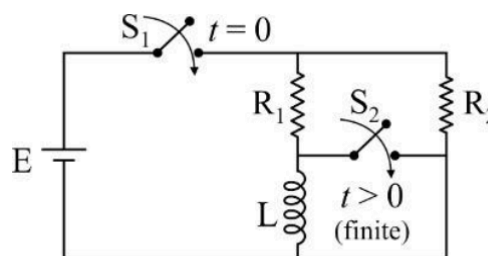
(A) $T_N = 20^\circ\text{C}$ (B) $T_M = 27.3^\circ\text{C}$ (C) $T_N = 16.4^\circ\text{C}$ (D) $T_M = 24^\circ\text{C}$

3. A small squirrel is put into a circular wheel cage of radius R which has a frictionless central pivot. A horizontal platform of length $\frac{6R}{5}$ is fixed to the wheel below the pivot as shown. Initially squirrel is at rest at one end of the platform. When the platform is released squirrel starts running but platform and wheel remain stationary. Choose the correct options.

- (A) Maximum speed of squirrel is $\sqrt{\frac{Rg}{5}}$
 (B) Maximum speed of squirrel is $\sqrt{\frac{9Rg}{20}}$
 (C) Maximum acceleration of squirrel is $\frac{4g}{3}$
 (D) Maximum acceleration of squirrel is $\frac{3g}{4}$

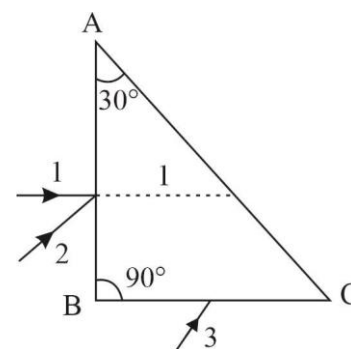


4. In the circuit shown below, the current through inductor is zero when battery is not connected ($t < 0$). At $t = 0$, switch S_1 is closed. If at a later time $t = t_1$ (finite), switch S_2 is also closed, then which of the following is true just after time t_1 ?



- (A) Current through resistor R_1 increases
(B) Current through resistor R_2 increases
(C) Current through resistor L increases
(D) Current through the battery increases

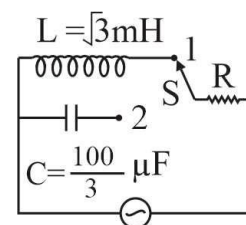
5. ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC . Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC . A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) have the same wavelength, then :



- (A) Refractive index of the material of prism is 2
(B) Angle of incidence in degree of ray (2) is 90
(C) Deviation in degree suffered by ray (2) is 60
(D) Deviation in degree suffered by ray (3) is 120

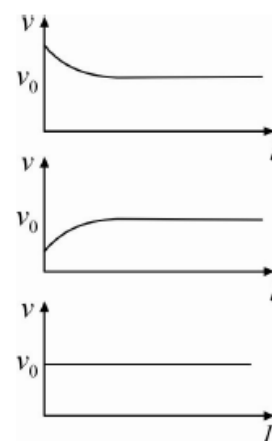
6. In the given AC circuit, when switch S is at position 1, the source e.m.f. leads current by $\frac{\pi}{6}$. Now, if the switch is at position 2, then current leads source e.m.f. by $\pi/4$, then :

- (A) Frequency of a.c source is $\frac{\sqrt{2.5}}{\pi} \text{ kHz}$
(B) $R = 3\sqrt{10} \Omega$
(C) Impedance of the circuit in position 1 is $2R/3$
(D) Impedance of the circuit in position 2 is $\sqrt{2}R$



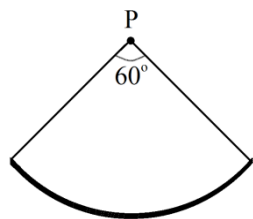
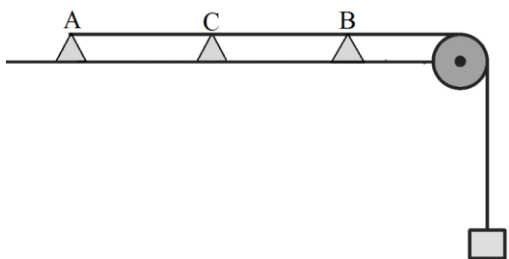
7. A student drops several small identical stones from different heights over a deep lake and prepares graphs between speed v of every stone in the water and time t . The graphs can be divided into three categories as shown here. Which of the following explanations of these graphs appear reasonable?

- (A) The first graph is for a stone dropped from a small height, and the second is for a stone dropped from a large height
(B) The first graph is for a stone dropped from a large height, and the second is for a stone dropped from a small height
(C) The third graph is for a stone dropped from a height sufficient to acquire the speed v_0 at the instant it enters the water
(D) The third graph is not possible



SECTION 2**NUMERICAL VALUE TYPE QUESTIONS**

This section consists of 5 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08).

8. The radii of two soap bubbles are 2cm and 3cm. They meet to produce a double bubble. The radius of their common interface in cm is _____.
9. A radioactive nuclei A decays into a stable nuclei B with half life 750 years. In an old sample ratio of number of nuclei of A to the number of nuclei of B is 1:7. Age of the sample is 500 n years. Find the value of n. Assume that initially equal number of nuclei of A and B were present in the sample.
10. A thin, uniform ring of radius $\frac{\pi}{4}$ m and mass $8\sqrt{3}$ kg is fixed with two rods of length $\frac{\pi}{4}$ m each. The system is pivoted about the centre of the ring P and made to rotate in the vertical plane. When the ring is in its lowest position (i.e. the position shown in the figure), the angular velocity of the system is 4 rad/s. At this instant, the force exerted by one of the rods on the ring is _____ (in N) [Assume that the force on the ring by a rod is always purely along the length of the rod, take $g = 10 \text{ m/s}^2$]
- 
11. A boy stands and looks at a small source of light on a wall at a distance 360 cm from him. The source is at the same height as his eyes, and his line of sight is perpendicular to the wall. Now, a convex lens of focal length 60 cm is introduced exactly midway between his eyes and the source of light such that the principal axis of the lens is coincident with his line of sight. Assuming that the size of an object perceived by a person is proportional to $\frac{1}{r}$, where r is the distance between the eyes of the person and the object, the number of times the size of the source of light as perceived by the boy increases due to the introduction of the lens is _____.
12. A thin metallic wire of mass per unit length 0.02kg/m is clamped at three points A, B and C and a block of weight 50 N is suspended from the wire as shown. The length AC = 25 cm and the length CB = 26 cm. If both segments (AC and CB) of the wire are made to vibrate in their fundamental modes, the number of beats heard per second (to the nearest integer) is _____.
- 

SECTION 3**MATRIX MATCH TYPE**

This Section has **2 Tables** (each having 3 columns and 4 rows). Based on each table, there are **Three (03) Matrix Match Type** questions. Each question has **FOUR** options out of which **ONLY ONE** is correct.

Table : Answer by appropriately matching the information given in the three columns of the following table.

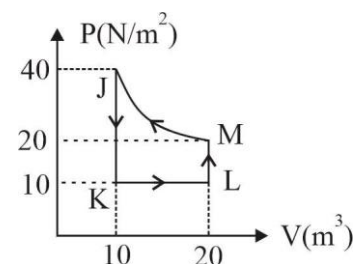
A charged particle (electron or proton) is introduced at the origin ($x = 0, y = 0, z = 0$) with a given initial velocity $\vec{v}$. A uniform electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in columns 1, 2 and 3, respectively. The quantities E_0, B_0 are positive in magnitude. ($v_0 \neq 0$)

Column 1		Column 2		Column 3	
(I)	Electron with $\vec{v} = \frac{E_0}{B_0} \hat{x}$	(i)	$\vec{E} = -E_0 \hat{z}$	(P)	$\vec{B} = -B_0 \hat{x}$
(II)	Electron with $\vec{v} = v_0 \hat{y} + v_0 \hat{x}$	(ii)	$\vec{E} = -E_0 \hat{y}$	(Q)	$\vec{B} = B_0 \hat{x}$
(III)	Proton with $\vec{v} = \vec{0}$	(iii)	$\vec{E} = -E_0 \hat{x}$	(R)	$\vec{B} = B_0 \hat{y}$
(IV)	Proton with $\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$	(iv)	$\vec{E} = \vec{0}$	(S)	$\vec{B} = \vec{0}$

13. In which case will the particle move in a straight line with constant velocity ?
(A) (I) (iii) (S) **(B)** (I) (i) (R) **(C)** (III) (iv) (R) **(D)** (III) (iii) (R)
14. In which case will the particle describe a helical path with constant pitch ?
(A) (II) (ii) (R) **(B)** (II) (iv) (P) **(C)** (IV) (i) (S) **(D)** (III) (iii) (P)
15. In which case would be particle move in a cycloid path ?
(A) (IV) (ii) (S) **(B)** (III) (i) (R) **(C)** (II) (iii) (Q) **(D)** (III) (ii) (R)

Table : Answer by appropriately matching the information given in the three columns of the following table.

An ideal gas is undergoing a cyclic thermodynamic process in different ways as shown in the corresponding P-V diagrams shown. Q denotes the heat supplied to the gas. W denotes the corresponding work done on the system. The equations and plots in the table have standard notations as used in thermodynamic processes. Here γ is the ratio of heat capacities at constant pressure and constant volume. The number of moles in the gas is n .



Column 1		Column 2		Column 3	
(I)	Process J-K	(i)	Isothermal	(P)	$W > 0$
(II)	Process K-L	(ii)	Isochoric	(Q)	$W < 0$
(III)	Process L-M	(iii)	Isobaric	(R)	$Q > 0$
(IV)	Process M-J	(iv)	Adiabatic	(S)	$Q < 0$

16. Which of the following options is the only correct representation of a process in which $\Delta U < 0$ & $\Delta U = Q$?

(A) (I) (iv) (R) (B) (II) (iii) (P) (C) (I) (ii) (S) (D) (III) (iii) (P)

17. Which one of the following options is the correct combination ?

(A) (III) (ii) (S) (B) (II) (iii) (R) (C) (II) (iv) (P) (D) (IV) (ii) (S)

18. Which one of the following options correctly represents a thermodynamic process that was used by Newton in the determination of the speed of sound in an ideal gas ?

(A) (III) (iv) (R) (B) (I) (ii) (R) (C) (IV) (i) (P) (D) (I) (iv) (Q)

SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY

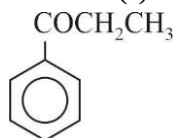
61 MARKS

SECTION 1

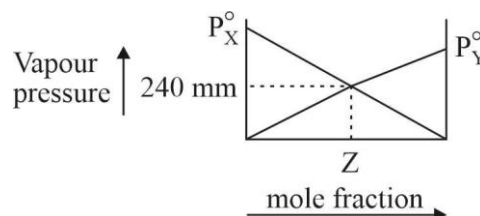
MULTIPLE CORRECT ANSWERS TYPE

This section consists of SEVEN (07) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. The correct name(s) of the following compound is(are) :



- (A) 1-phenyl-1-propanone (B) 3-phenyl-3-propanone
(C) ethyl phenyl ketone (D) methyl phenyl ketone
2. The correct statement(s) for the following reactions is (are):
- (i) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow{\text{H}_2 / \text{Pd} / \text{CaCO}_3} [\text{M}] \xrightarrow{\text{Br}_2 / \text{CCl}_4} [\text{N}]$
- (ii) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow{\text{Br}_2 / \text{CCl}_4} [\text{O}] \xrightarrow{\text{H}_2 / \text{Pd}} [\text{P}]$
- (A) Products P and N are identical
(B) M will be the cis-alkene where as O will be the trans-alkene
(C) (P and N) and (M and O) are pair of enantiomers
(D) (P and N) and (M and O) are pair of diastereomers
3. $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ dissolve in water to form a complex. If a solution prepared by dissolving 0.1 mole of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ in water is treated with excess AgNO_3 solution, 0.2 mole of AgCl is precipitated. Select correct option for complex : (Mass number of Cr = 52, O = 16, Cl = 35.5)
- (A) It will exist in hydrate isomerism
(B) It loses almost 6.75% of its original mass on treatment with conc. H_2SO_4
(C) It is a paramagnetic complex in which central metal is in d^2sp^3 hybridisation
(D) It will exist in optical isomeric form
4. Two volatile liquids X and Y form an ideal binary solution and variation of vapour pressure versus mole fraction is described in the following diagram. Which of the following statement regarding pure X, pure Y and the solution of X and Y is correct ?

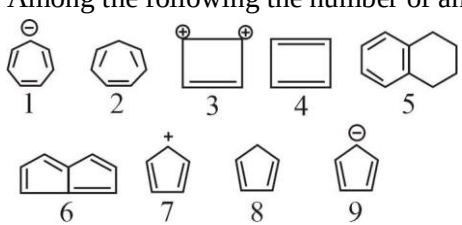


5. An ideal gas is taken reversibly from state A (P, V) to the state B (0.5P, 2V) along a straight line in PV diagram. Which of the following statement(s) is/are correct regarding the process ?
- (A) The work done by gas in process A to B exceeds the work that would be done by it if the same change in state were performed isothermally
- (B) In the T-V diagram, the path AB becomes a parabola
- (C) In the P-T diagram, the path AB becomes a hyperbola
- (D) On going from A to B, the temperature of the gas first increases to a maximum value and then decreases
6. The correct statement(s) about the oxoacids $\text{H}_5\text{P}_5\text{O}_{15}$ and $\text{H}_7\text{P}_5\text{O}_{16}$ is(are) :
- (A) Both have equal number of sp^3 hybridised 'P' atoms
- (B) Both have same basicity
- (C) Both have different number of $\text{p}\pi - \text{d}\pi$ linkages
- (D) Both have unequal number of P – O – P linkages
7. Liquid oxygen which belongs to group 16 is blue in colour due to :
- (A) Electron transition from HOMO to LUMO in visible region
- (B) Electron transition from HOMO to LUMO in ultra-violet region
- (C) It contains 2 unpaired electrons which create magnetic asymmetry
- (D) Decrease in ionization enthalpy

SECTION 2

NUMERICAL VALUE TYPE QUESTIONS

This section consists of 5 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08).

8. Among B_2 , C_2^{2-} , N_2^+ , O_2^- , NO^+ , OF , find the total number of species which is/are paramagnetic and have fractional bond order.
[Atomic Number : B = 5, C = 6, N = 7, O = 8, F = 9]
9. Among the following the number of anti-aromatic compounds is/are:
- 
10. Molar conductivity of 0.04 M MgCl_2 solution at 298K is $200 \text{ S cm}^2 \text{ mole}^{-1}$. A conductivity cell which is filled with MgCl_2 have area of cross-section of electrode 4 cm^2 & distance between electrode is 8cm. If potential difference between electrode is 10V then find current flow in mA.
(Report your answer after dividing by 10)

11. The sum of the number of lone pair of electrons on each central atom in the following species is :
 $[\text{XeF}_5]^-$, TeBr_4 , $[\text{IO}_2\text{F}_2]^-$, SeOF_2
 [Atomic Number : Xe = 54, F = 9, Te = 52, Br = 35, I = 53, O = 8, Se = 34]
12. A metallic element exists as cubic lattice. Edge length of the unit cell is $2\text{\AA}$. If the density of the metal is 6.25 g/cm^3 , then the number of unit cells present in 100g of the metal is $x \times 10^{24}$. The value of x is :

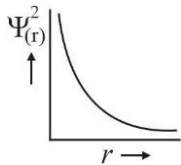
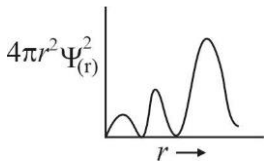
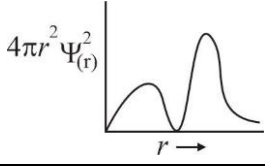
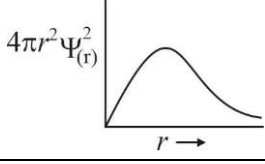
SECTION 3

MATRIX MATCH TYPE

This Section has **2 Tables** (each having 3 columns and 4 rows). Based on each table, there are **Three (03) Matrix Match Type** questions. Each question has **FOUR** options out of which **ONLY ONE** is correct.

Table-1 : Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table.

The wave function Ψ_{n, ℓ, m_ℓ} is a mathematical function whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterized by the quantum numbers n, ℓ and m_ℓ . Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the table, z is atomic number and a_0 is Bohr radius.

Column 1		Column 2		Column 3	
(I)	3d	(i)	Two nodal plane	(P)	
(II)	3p	(ii)	$\Psi_{n, \ell, m_\ell} \propto \left(\frac{z}{a_0}\right)^{3/2} e^{\left(-\frac{zr}{a_0}\right)}$	(Q)	
(III)	3s	(iii)	One radial node and one nodal plane	(R)	
(IV)	1s	(iv)	probability density is maximum at nucleus	(S)	

13. Which of the following is correct combination for outermost subshell of Cu^{2+} ion ?
 (A) (II) (i) (Q) (B) (I) (i) (Q) (C) (I) (i) (S) (D) (I) (iii) (S)
14. An electron is in excited state in hydrogen atom. The orbital in which electron present is 3s. Select correct combination for this :
 (A) (III) (ii) (P) (B) (III) (iv) (Q) (C) (III) (iv) (R) (D) (III) (iii) (S)
15. For Li^{2+} ion, the only correct combination is :
 (A) (IV) (ii) (P) (B) (IV) (iv) (R) (C) (III) (iv) (P) (D) (III) (ii) (P)

Table-2 : Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contains starting materials reaction conditions and type of reactions respectively.

Column 1		Column 2		Column 3	
(I)	Benzene	(i)	$h\nu / \text{Cl}_2$	(P)	Acylation
(II)	Ethanal	(ii)	$\text{RCOCl} / \text{anhydrous AlCl}_3$	(Q)	Free radical addition
(III)	Toluene	(iii)	dil. NaOH	(R)	Allylic Substitution
(IV)	Ethyl Benzene	(iv)	NBS	(S)	Condensation

16. For the synthesis of aromatic ketones, the only correct combination is :

(A) (III) (i) (R) (B) (I) (ii) (P) (C) (I) (iv) (Q) (D) (II) (i) (S)

17. The only correct combination that gives a product (can be intermediate product) having both alcohol and aldehyde groups.

(A) (II) (iii) (S) (B) (I) (ii) (P) (C) (I) (iv) (Q) (D) (I) (i) (S)

18. The only correct combination in which the reactions proceeds through free radical mechanism :

(A) (I) (i) (R) (B) (III) (i) (P) (C) (IV) (iv) (R) (D) (III) (i) (S)

SPACE FOR ROUGH WORK

SECTION 1

MULTIPLE CORRECT ANSWERS TYPE

This section consists of SEVEN (07) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

- If A and B are two events such that $P(A) = 1/2$ and $P(B) = 2/3$, then $P(A' \cap B)$ may be :
 (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{5}$
- The equation of the curve such that the distance between the origin & the tangent line at an arbitrary point is equal to the distance between the origin and the normal at the same point is :
 (A) $x^2 + y^2 = c \cdot e^{\tan^{-1}(y/x)}$ (B) $\sqrt{x^2 + y^2} = c \cdot e^{\tan^{-1}(y/x)}$
 (C) $\sqrt{x^2 + y^2} = c \cdot e^{-\tan^{-1}(y/x)}$ (D) $x^2 + y^2 = c \cdot e^{-\tan^{-1}(y/x)}$
 Where c is an arbitrary constant ?
- If from a point P representing the complex number z_1 on the curve $|z| = 2$, pair of tangents are drawn to the curve $|z| = 1$, meeting at point Q(z_2) and R(z_3), then :
 (A) complex number $\frac{z_1 + z_2 + z_3}{3}$ will lie on the curve $|z| = 1$
 (B) $\left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3}\right)\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right) = 9$
 (C) $\left|\arg\left(\frac{z_2}{z_3}\right)\right| = \frac{2\pi}{3}$
 (D) orthocentre and circumcentre of ΔPQR will coincide
- The equation(s) to common tangent(s) to the two hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ is/are :
 (A) $y = x + \sqrt{a^2 - b^2}$ (B) $y = x - \sqrt{a^2 - b^2}$
 (C) $y = -x + \sqrt{a^2 - b^2}$ (D) $y = -x - \sqrt{a^2 - b^2}$
- $f(x) = \sin^{-1} \frac{2x}{1+x^2}$, then :
 (A) f is derivable for all x, with $|x| < 1$
 (B) f is not derivable at $x = 1$
 (C) f is not derivable at $x = -1$
 (D) f is derivable for all x, with $|x| > 1$
- A be set of 3×3 matrices formed by entries 0, -1, and 1 only. Also each of 1, -1, 0 occurs exactly three times in each matrix. The number of symmetric matrices with trace (A) = 0 is :
 (A) 36 (B) 30 (C) 28 (D) 34

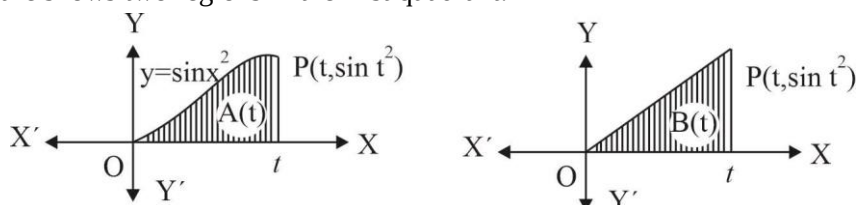
7. If $\cos(\alpha - \beta)$ and $\cos(\alpha + \beta)$ are length of segments of a focal chord of parabola $y^2 = 2\cos\alpha x$ divided at axis, then $\cos\alpha \sec\frac{\beta}{2} (\cos\beta \neq 1)$ is equal to :
- (A) 1 (B) -2 (C) $\sqrt{2}$ (D) $-\sqrt{2}$

SECTION 2

NUMERICAL VALUE TYPE QUESTIONS

This section consists of 5 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08).

8. Let $A_n, (n \in \mathbb{N})$ be a matrix of order $(2n-1) \times (2n-1)$, such that $a_{ij} = 0 \forall i \neq j$ and $a_{ij} = n^2 + i + 1 - 2n \forall i = j$ where a_{ij} denotes the element of i^{th} row and j^{th} column of A_n . Let $T_n = (-1)^n \times (\text{sum of all the elements of } A_n)$. Find the value of $\left\lfloor \frac{\sum_{n=1}^{102} T_n}{520200} \right\rfloor$, where $[.]$ represents the greatest integer function.
9. For any odd integer $n \geq 1$, if the value of the sum : $n^3 - (n-1)^3 + (n-2)^3 + \dots + (-1)^{n-1} (1^3)$ equals 208 then the value of n is equal to :
10. The figure shows two regions in the first quadrant.



$A(t)$ is the area under the curve $y = \sin^2 x$ from 0 to t and $B(t)$ is the area of the triangle with vertices O, P and $M(t, 0)$. Then the value of $\lim_{t \rightarrow 0} \frac{12A(t)}{B(t)}$ is :

11. Let the area of the circle which passes through $(2, 0)$ and whose centre is the limit of the point of intersection of the lines $3x + 5y = 1$ and $(2+c)x + 5c^2y = 1$ as $c \rightarrow 1$ is $\frac{\pi a}{625}$, then find a .
12. Let A be a set containing 9 elements and B be a set containing 3 elements. Let N = Number of onto functions from $A \rightarrow B$. Then $\left\lfloor \frac{N}{11} \right\rfloor$ = (Where $[]$ denotes represents the greatest integer function)

SECTION 3**MATRIX MATCH TYPE**

This Section has **2 Tables** (each having 3 columns and 4 rows). Based on each table, there are **Three (03) Matrix Match Type** questions. Each question has **FOUR** options out of which **ONLY ONE** is correct.

Table-1 : Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3 (If $a = 2$, then curve passes through)	
(A)	A tangent to the parabola $x^2 + 4ay = 0$ cuts the parabola $x^2 = 4y$ at A and B. The locus of mid-point of AB is	(P)	$y^2(x + 2a) + 4a^3 = 0$	(1)	$(-4, -2)$
(B)	Locus of the mid point of the chords of the hyperbola $x^2 - y^2 = b^2$ of slope 'a'	(Q)	$x = ay$	(2)	$(-6, -4)$
(C)	Locus of the points of intersection of tangents drawn at the ends of all normal chords of the parabola $y^2 = 4ax$	(R)	$(a + 2)x^2 = 4y$	(3)	$(-1, 1)$
(D)	Locus of the foot of the perpendicular from the centre of the hyperbola $xy = a^2$ on a variable tangent is	(S)	$(x^2 + y^2)^2 = 4a^2xy$	(4)	$(0, 0)$

13. Correct sequence among the following is :

- (A) BQ2 (B) BQ4 (C) DS2 (D) DQ4

14. Correct sequence among the following is :

- (A) AR4 (B) AR1 (C) AQ2 (D) DR4

15. Correct sequence among the following is :

- (A) CS4 (B) CP3 (C) DS4 (D) CR3

Table-2 : Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

Column I		Column II $k =$		Column III $l =$	
(A)	$k =$ Number of integral values of x satisfying $ x^2 - 9 + x^2 - 4 = 5$ $l =$ Largest value of x satisfying $ x^2 - 9 + x^2 - 4 = 5$	(P)	0	(1)	0
(B)	Let $f(x) = \begin{cases} -5, & -\infty < x \leq -1 \\ x-4 & -1 < x \leq 6, \\ 2, & 6 < x < \infty \end{cases}$ then Let n be the number of integers in the range of $f(x)$ $k = n - 4$ and $l = \lim_{x \rightarrow n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)}$ where $[x]$ denotes greatest integer less than or equal to x .	(Q)	1	(2)	1
(C)	If $\lim_{x \rightarrow 0} \frac{a \sin x + bxe^x + 3x^2}{\sin x - 2x + \tan x}$ exists and has value equal to L . $k = a + b$ $l = \left\lfloor \frac{L}{3} \right\rfloor$	(R)	2	(3)	2
(D)	Let $P_n = \prod_{m=2}^n \left(1 - \frac{1}{m+1} C_2\right)$. $\lim_{n \rightarrow \infty} P_n = \frac{k}{l}$ $\{k, l \in N \text{ and they are co-prime}\}$	(S)	3	(4)	3
		(T)	4	(5)	4

16. Correct sequence among the following is :

- (A) AP4 (B) BQ3 (C) CP5 (D) DQ2

17. Correct sequence among the following is :

- (A) BT1 (B) CQ2 (C) DS3 (D) AT4

18. Correct sequence among the following is :

- (A) DQ4 (B) DQ5 (C) DR4 (D) DS5

SPACE FOR ROUGH WORK

